



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE20CS353 - Compiler Design

Total Marks : 100.0

1.a. i) What is Interpreter? How it is different from a Compiler.
ii) Explain in brief (any two) types of compilers. (4.0 Marks)

1.b. Explain in brief the role of a lexer. Is it possible to have a common lexer for all the languages? Justify your answer with at least 2 examples. (4.0 Marks)

1.c. With a diagram explain in detail the design of a lexical analyzer. (4.0 Marks)

1.d. What are Lexical Errors? Write down the sequence of lexemes for the following code segment
`while(i <= 10) { printf("%d",i); i++; }` (4.0 Marks)

1.e. i) What is Lex tool? Mention the sections of Lex specification program.
ii) Consider the following lexical specification:

```
%%  
a*b           { printf(" 1 "); }  
(a|b)*b      { printf(" 2 "); }  
c*           { printf(" 3 "); }
```

Explain how lex would tokenize the following input string and the output produced.

(I) cbbbbac (II) cbabc (4.0 Marks)

2.a. Explain in detail any two Error-Recovery Strategies. (4.0 Marks)

2.b. Consider the following grammar:

$S \rightarrow +SS \mid *SS \mid a$

1. Calculate FIRST and FOLLOW
2. Construct predictive parsing table
3. Show how to parse the input $+*aaa$

(6.0 Marks)

2.c. Consider the following grammar:

$S \rightarrow SS+ \mid SS* \mid a$

1. Construct LR(0) automata
2. Construct SLR parsing table to check whether the given grammar is SLR or not.

(6.0 Marks)

2.d. i) Explain in brief Canonical LR(1) Items.

ii) Why is LR(1)/CLR(1) parser so powerful?

(4.0 Marks)

3.a. Explain the following

i) Synthesized and Inherited Attributes

ii) L-attributed SDT and S-attributed SDT

(8.0 Marks)

3.b. Consider the following grammar for variable declaration in C language:

$D \rightarrow TL$

$T \rightarrow \text{int} \mid \text{float} \mid \text{char} \mid \text{double}$

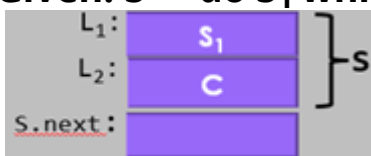
$L \rightarrow L_1, \text{id} \mid \text{id}$

Write semantic rules to add/update type(int/float/char/double) information of the variables in declaration statement to the symbol table during parsing. (4.0 Marks)

3.c. Write syntax directed definition (SDD) to generate intermediate code for do-while loop.

Given: $S \rightarrow \text{do } S_1 \text{ while}(C)$

(4.0 Marks)



3.d. Write syntax directed definition (SDD) to generate three-address intermediate code for given Conditional Boolean expression production.

$$C \rightarrow B_1 \mid \mid B_2$$

Note: Use function *new()* that generates new labels. (4.0 Marks)

4.a. Write short notes on:

i) Quadruple

ii) Static Single Assignment (SSA)

(5.0 Marks)

4.b. Construct control-flow graph (CFG) for the following three-address code

(5.0 Marks)

```
x = 0
y = 0
L1: ifFalse x < 10 goto L2
    t1 = y + x;
    y = t1
    t2 = x + 1;
    x = t2
    goto L1
L2: print y
```

4.c. Optimize the code below by applying the following code transformations: Constant Folding, Constant Propagation and dead code elimination.

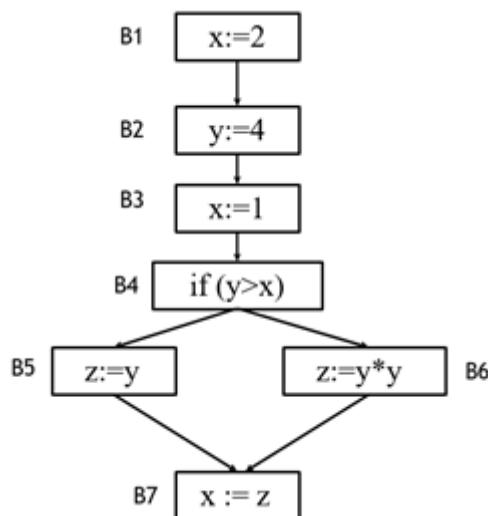
(5.0 Marks)

```
int a = 30;
int b = 9 - (a / 5);
int c;
c = b * 4;

if (c > 10) {
  c = c - 10;
}
return c * (60 / a);
```

4.d. Calculate use[B], def[B] and then find IN[B], OUT[B] by applying *Live-variable Analysis Algorithm* on the following flow graph.

(5.0 Marks)



5.a. Explain in brief any four issues in the design of a code generator.

(4.0 Marks)

5.b. Explain the following with a neat diagram

i) Activation record

ii) Activation tree

(6.0 Marks)

5.c. Explain in brief the different ways in which a procedure can access nonlocal data on a runtime Stack.

(4.0 Marks)

5.d. Generate target code sequence for the following basic block three address statements with register and address descriptors.

1. $x := y - z$

2. $z := x * x$

3. $y := z$

4. $x := y - z$

Note 1: Assuming only two registers (R1 and R2) are available and all the variables are live on exit from the basic block. No temporary variables are used.

Note 2: Assuming that all arithmetic operations take their operands from registers. (6.0 Marks)